

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 8

For the tutorial in Week 9 on 21 March, let us discuss the following questions.

Question 1. Draw and determine whether each of these graphs is strongly connected and if not, whether it is weakly connected.

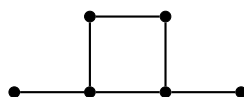
- (a) $G = (V, E)$ with $V = \{a, b, c, d, e\}$, and directed edges
 $E = \{(b, a), (b, c), (b, e), (e, a), (d, b), (c, d)\}$.
- (b) $G = (V, E)$ with $V = \{a, b, c, d, e\}$, and directed edges
 $E = \{(a, d), (a, e), (b, a), (b, c), (d, b), (d, c), (e, b), (e, d)\}$.
- (c) $G = (V, E)$ with $V = \{a, b, c, d, e, f, g\}$, and directed edges
 $E = \{(a, c), (a, g), (g, d), (d, c), (e, b), (b, f), (f, e)\}$.

Solution:

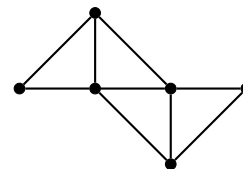
- (a) Weakly connected, e.g., there is no path from a to b.
- (b) Weakly connected, e.g., there is no path from c to b.
- (c) Not connected. There is no path between a and b in the corresponding undirected graph.

Question 2. The *degree sequence* of a graph $G = (V, E)$ is the sequence of length $n = |V|$ that contains $\deg(v)$ for all $v \in V$. Usually it is sorted non-decreasingly.

- (a) State the degree sequence of the following graphs:



Graph 1



Graph 2

- (b) Prove that two graphs that do not have the same degree sequence cannot be isomorphic.

Solution:

- (a) The degree sequence of graph 1 is $(1, 1, 2, 2, 3, 3)$. The degree sequence of graph 2 is $(2, 2, 3, 3, 4, 4)$. Notice the even number of odd-degree vertices.
- (b) We prove the contrapositive statement, that is, if two graphs are isomorphic, then they have the same degree sequence.

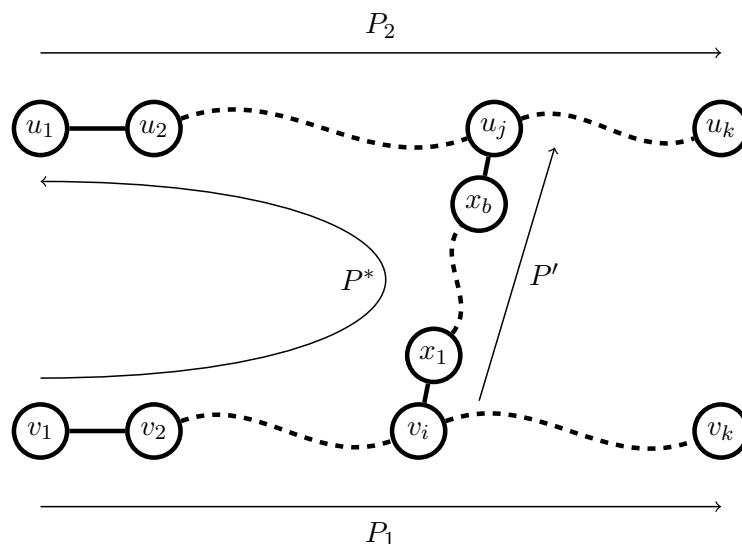
Suppose $G = (V_1, E_1)$ and $H = (V_2, E_2)$ are isomorphic. Then there is a mapping $f : V_1 \rightarrow V_2$ such that $(u, v) \in E_1 \Leftrightarrow (f(u), f(v)) \in E_2$. This implies that v must have the same degree as $f(v)$: for every edge (u, v) there is an edge $(f(u), f(v))$ and vice versa. Then the degree sequence of G and H must also be the same.

Question 3. If a graph G is connected and has no simple path with length greater than k , prove that every two simple paths in G of length k have at least one vertex in common.

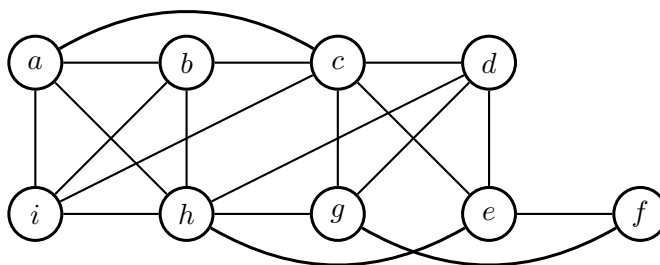
Solution: Suppose not. That is, suppose we have two paths, $P_1 = (v_1, \dots, v_k)$ and $P_2 = (u_1, \dots, u_k)$, in G of length k with no shared vertices.

Since G is connected, there is a path P' connecting v_i to u_j for some $i, j \in [1, k]$ such that P' shares no vertices with $P_1 \cup P_2$ other than v_i and u_j . Say $P' = (v_i, x_1, \dots, x_b, u_j)$ (note that there may be no x_i vertices, i.e., $P' = (v_i, u_j)$).

Without loss of generality we may assume that $i, j > \lceil \frac{k}{2} \rceil$ (we can always reverse the numbering). Then we can construct a new path $P^* = (v_1, \dots, v_i, x_1, \dots, x_b, u_j, \dots, u_1)$ (by going along P_1 to v_i , then across the bridge formed by P' to u_j , then along P_2 to u_1). Obviously P^* has length at least $k + 1$, but this contradicts the assumption that G has no paths of length greater than k .



Question 4. (Ex. 10.5.4.) Does this graph have an Euler circuit? Find one if there is one.



Solution: It is easy to check that every vertex has even degree, hence there is an Euler circuit.

To find one we start at vertex a and create a circuit by following edges.

One such circuit is

$$(a, c, g, h, i, a, b, h, a).$$

On this circuit b, c, g, i still have unused edges. We start from i and get the circuit

$$(i, b, c, d, e, c, i).$$

The two circuits can be joined into

$$(a, c, g, h, i, b, c, d, e, c, i, a, b, h, a).$$

On this circuit d, h, g, e have unused edges. We start from h and get a new circuit:

$$(h, d, g, f, e, h).$$

The circuits can be joined to form

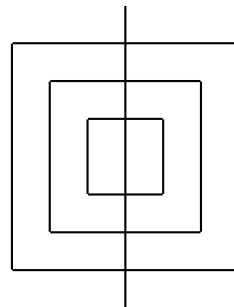
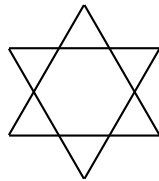
$$(a, c, g, h, i, b, c, d, e, c, i, a, b, h, d, g, f, e, h, a)$$

which is an Euler circuit.

Question 5. Is it true that by adding edges to a simple graph $G = (V, E)$ (without introducing multiple edges between any pair of vertices) one can always create a graph that has an Euler circuit? Justify the answer.

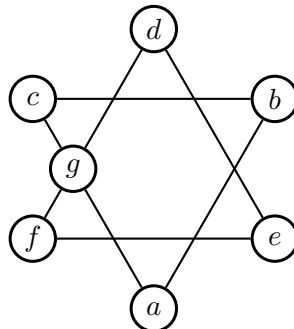
Solution: The graph K_4 has 4 vertices of odd degree, and it is not possible to add more edges without having multiple edges between vertices. Hence this is not possible.

Question 6. (Ex. 10.5.7, 8.) Determine whether each of the picture shown below can be drawn with a pencil in a continuous motion without lifting the pencil or retracing part of the picture.



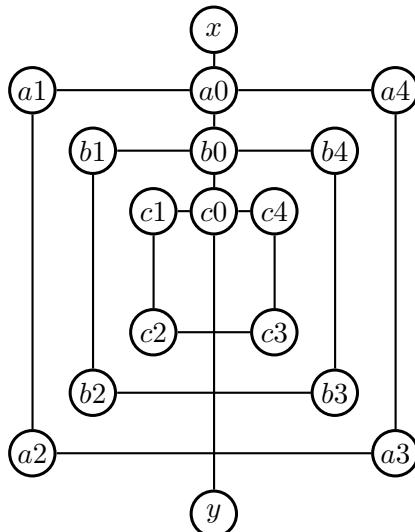
Solution:

The first picture can be represented using the edges of the following graph.



We can easily check that the degrees at each vertex is even, and so an Euler circuit exist, for example, $(g, a, b, c, g, d, e, f, g)$. Then by following the directions of this Euler circuit, we may draw the picture with a pencil in a continuous motion without lifting the pencil or retracing part of the picture.

Similarly, the second picture can be represented using the edges of the following graph.



We can easily check that the degree of the vertices x and y are odd while the degrees at every other vertex is even, and so an Euler path exist, for example,

$$(x, a0, a1, a2, a3, a4, a0, b0, b1, b2, b3, b4, b0, c0, c1, c2, c3, c4, c0, y).$$

Then by following the directions of this Euler path, we may draw the picture with a pencil in a continuous motion without lifting the pencil or retracing part of the picture.

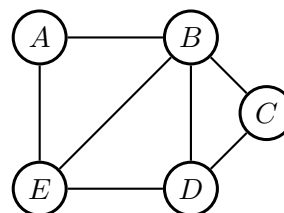
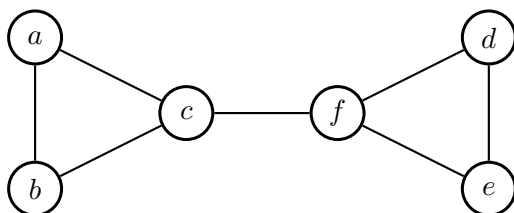
Question 7 (Ex. 10.5.16). For what values of n do these graphs have an Euler circuit?

- (a) K_n ,
- (b) C_n .

Solution:

- (a) For K_n to contain a circuit, we need n at least 3. Note that every vertex in K_n has degree $n - 1$, and in order to have an Euler circuit, we need the degree at every vertex to be even. Therefore, K_n has an Euler circuit for any odd $n \geq 3$.
- (b) Since the degree of any vertex in C_n is 2, all C_n have Euler circuits for $n \geq 3$.

Question 8 (Ex. 10.5.20, 21). Determine whether each of the given graph below has a Hamilton circuit. If it does, find such a circuit. If it does not, give an argument to show why no such circuit exists.



Solution: A Hamilton circuit does not exist for the first graph. Any Hamilton circuit for the first graph must contain the edge $\{c, f\}$ twice.

Yes, a Hamilton circuit exist for the second graph. We may simply use $A-B-C-D-E-A$.

The following exercise will not be discussed during tutorial. The inequality $n - k \leq m$ is manageable but the other inequality $m \leq (n - k)(n - k + 1)/2$ is quite involved. You may ignore this question for test and exam purposes.

Question 9. Given a simple graph G with n vertices and m edges. If G has exactly k connected components, prove that

$$n - k \leq m \leq \frac{(n - k)(n - k + 1)}{2}.$$

Solution: Assume the number of vertices in each component C_i is x_i , where $x_1 + \dots + x_k = n$, and x_i are positive integers for all i .

First we show $n - k \leq m$. To show this, we first prove that a connected graph G with x vertices always has at least $x - 1$ edges. We start from an arbitrary vertex $v_1 \in G$. Since the graph is connected, there is an edge, say $\{v_1, v_2\}$, connecting v_1 to $G - \{v_1\}$. Then there is an edge, say $\{v_2, v_3\}$, connecting v_2 to $G - \{v_1, v_2\}$. Repeat this process until we have all x vertices included, and we have found $x - 1$ different edges along the process.

Having this claim, it is now straightforward to see that if the graph G has k connected components, we have $m \geq (x_1 - 1) + (x_2 - 1) + \dots + (x_k - 1) = \sum_i x_i - k = n - k$.

Next we show $m \leq \frac{(n-k)(n-k+1)}{2}$. For each connected component C_i with x_i vertices, the number of edges inside is at most $\frac{x_i(x_i-1)}{2}$. Thus

$$m \leq \sum_i \frac{x_i(x_i - 1)}{2} = \frac{1}{2} \sum_i x_i^2 - \frac{1}{2} \sum_i x_i = \frac{1}{2} \sum_i x_i^2 - \frac{n}{2}.$$

Now we find the maximum value of $\sum_i x_i^2$. If there are two elements x_i and x_j that are both strictly greater than 1, assume without loss of generality that $x_i \leq x_j$. Then consider $x'_i = x_i - 1$ and $x'_j = x_j + 1$, we have $x'^2_i + x'^2_j = (x_i - 1)^2 + (x_j + 1)^2 = x^2_i + x^2_j + 2 + 2(x_j - x_i) > x^2_i + x^2_j$. This means when $\sum_i x_i^2$ is maximized, at most one x_i is strictly greater than 1, and the rest must all be 1 (i.e., isolated vertices). Thus

$$m \leq \frac{1}{2} \sum_i x_i^2 - \frac{n}{2} \leq \frac{(n - k + 1)^2 + k - 1}{2} - \frac{n}{2} = \frac{(n - k)(n - k + 1)}{2}.$$